

Comparing Malicious vs. Benign Twitter Subgraphs (WICO Dataset)

1. Introduction

- This report analyzes two Twitter subgraphs from the WICO dataset:
 1. One from 5G_Conspiracy_Graphs (misinformation cluster)
 2. One from Non_Conspiracy_Graphs (normal Twitter community)
- The goal is to compare their structural properties using Social Network Analysis (SNA) in Gephi and understand how misinformation networks differ from normal online communities.

2. Methodology

Both graphs were loaded into Gephi, visualized using the ForceAtlas2 layout, and analyzed using the following graph metrics:

- Number of Nodes & Edges
- Average Degree
- Graph Density
- Average Clustering Coefficient
- Modularity (Q) and number of communities
- Connected Components
- Network Diameter (Betweenness & Closeness included)
- Average Path Length

Outputs from Gephi are interpreted below.

3. Results

	5G_Conspiracy_Graph	Non_Conspiracy_Graph
Nodes	89	58
Edges	42	127
Average Degree	0.472	2.19
Network Diameter	4	7
Graph Density	0.005	0.038
Connected Components	62	8
Modularity	0.677	0.378
Avg. Clustering Coefficient	0.012	0.271
Avg. Path Length	1.764	2.981

In 5G_Conspiracy_Graph :

- The graph is extremely sparse, with very few connections between users.
- A very high number of connected components (62) indicates an extremely fragmented network.
- High modularity (0.677) shows strong echo chambers—small isolated groups discussing the same misinformation. (An echo chamber is a tightly connected community where users mainly interact with each other)
- Very low clustering coefficient indicates weak interaction inside groups.

Non_Conspiracy_Graph :

- The graph is highly connected with significantly more interactions.
- Only 8 connected components, meaning most users belong to larger, well-connected communities.
- Lower modularity shows open, less isolated communication.
- High clustering coefficient shows strong internal cohesion and natural social grouping.